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Paper 1: Net Renal Drug Secretion Critically Requires Significant Contribution of Basal U

Bibliographic Information

- **Title:** Net Renal Drug Secretion Critically Requires Significant Contribution of Basal Uptake Transporters: Perspective From Mechanistic Kidney Model Simulation and Integration of Clinical and In Vitro Data
- **Authors:** Tsuchitani T, Tomi M, Sugiyama Y
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2026
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Quick Take

This mechanistic simulation study defines the quantitative conditions required for net renal secretion ($CR > 1.5$), showing that basal uptake transporters (e.g., OAT1/3, OCT2) are essential. It provides in vitro-translatable thresholds (e.g., $>94\%$ transporter contribution to basal uptake) that can predict secretion and DDI risk. Read this if you are involved in renal clearance prediction, transporter DDI assessment, or PBPK modeling of renally cleared drugs.

Executive Summary

This study uses a mechanistic kidney model (MechKiM) to systematically define the quantitative requirements for net renal secretion (clearance ratio $CR > 1.5$). Simulations of 30,000 virtual drugs across all combinations of three proximal-tubule transporters (basal uptake, basal efflux, apical efflux) reveal that basal uptake transport is indispensable; apical efflux alone cannot produce net secretion because proximal secretion is offset by downstream reabsorption. Five universal thresholds are identified, including a $>94\%$ contribution of transporter-mediated uptake to total basal uptake and a minimum basal uptake clearance of ~ 4.2 L/h. These findings are corroborated by clinical data (70 of 110 secretory drugs are uptake-transporter positive) and published MechKiM-PBPK models (typically $>99\%$ basal uptake via transporters). The work provides practical, in vitro-translatable criteria for predicting renal secretion and DDI risk, advancing mechanistic renal clearance prediction.

Scientific Context & Motivation

Renal clearance is a composite of filtration, secretion, and reabsorption, making it difficult to predict net secretion from simple clearance models. While in vitro substrate recognition of OAT1/3 and OCT2 is a qualitative predictor, the quantitative kinetic requirements for net secretion ($CR > 1.5$) have been unclear. This study addresses the gap by systematically exploring the parameter space with a mechanistic kidney model, identifying universal thresholds for elementary processes

(passive diffusion, transporter clearances) that govern net secretion. It also reconciles conflicting observations from PBPK models (e.g., rivaroxaban) and provides a framework for translating in vitro data to in vivo secretion potential.

✂ Methodological Snapshot

The study employs a mechanistic kidney model (MechKiM) to simulate renal clearance for a large virtual drug library. The model includes glomerular filtration, passive diffusion with pH-dependent ionization, and transporter-mediated processes (basal uptake, basal efflux, apical efflux) localized to the proximal tubule. Steady-state $CL_{r,b}$ is computed for 30,000 virtual drugs across all transporter knockout combinations. Thresholds for net secretion are derived by analyzing parameter- $CL_{r,b}$ relationships. The findings are validated against clinical $CL_{r,p}$ data and published MechKiM-PBPK models.

☒ Structural Model Breakdown

The MechKiM model consists of four nephron segments: glomerulus, proximal tubule (PT), distal tubule (DT), and collecting duct (CD). Each tubular segment is represented by three compartments: blood (vascular space), epithelial cells, and lumen. The glomerulus has only blood and lumen compartments. Transporters are localized to the PT epithelial cells: basal uptake ($PS_{act,influx(B)}$), basal efflux ($PS_{act,efflux(B)}$), and apical efflux ($PS_{act,efflux(A)}$). Passive diffusion is described by permeability (P_d) across cell membranes, with pH-dependent ionization affecting the unionized fraction. The model includes blood flow (Q_u) and surface areas (SA) for each segment. The differential equations for PT, DT, and CD lumen concentrations are given in Equations 13–15. The model is connected to a central blood compartment, and steady-state $CL_{r,b}$ is simulated under continuous infusion.

Detailed Methodological Analysis

Modeling Approach

Mechanistic kidney model (MechKiM) with four nephron segments (proximal tubule, distal tubule, collecting duct) and three compartments per segment (blood, epithelial cell, lumen). Transporters (basal uptake, basal efflux, apical efflux) were placed only in proximal tubule cells. Passive diffusion was modeled with pH-dependent ionization and membrane potential effects. Virtual drugs were generated by sampling parameters (P_d , PS_{act} , LR_{renal} , pK_a) from log-uniform distributions.

Data Sources

Simulation study using MechKiM with 30,000 virtual drugs (10,000 parameter sets $\times$ 3 charge classes). Clinical $CL_{r,p}$ data for 516 drugs from published sources (Varma et al., Ito et al., and previous report) were filtered to 217 drugs with complete in vitro transporter data. Literature review identified 31 MechKiM-PBPK models covering 19 drugs.

Estimation Methods

Deterministic simulation of steady-state CL_{r,b} under continuous intravenous infusion. No parameter estimation was performed; all parameters were sampled from log-uniform distributions. The model was implemented in a mechanistic kidney model framework (MechKiM) with linear transporter kinetics.

Model Evaluation

Simulation results were compared with clinical data (in vitro transporter recognition vs. observed CL_r) and with published MechKiM-PBPK models. The consistency of thresholds across fu_b values and the robustness to assumptions (e.g., ionized permeability) were assessed via sensitivity analyses.

Covariate Analysis

The effects of fu_b (1 vs 0.01), charge class (neutral, anion, cation), and transporter knockout combinations were systematically evaluated. Thresholds were derived for each scenario, and sensitivity to fu_b was assessed. The impact of adding a distal tubule uptake transporter was also explored.

Statistical Rigor Assessment

The simulation design is robust, with 30,000 virtual drugs covering a wide parameter space (log-uniform sampling). The use of 10,000 independent parameter sets per charge class ensures comprehensive coverage. Thresholds were derived from clear boundaries in parameter–CL_r relationships, and sensitivity analyses (fu_b = 0.01, ionized permeability, distal tubule transporters) confirmed robustness. Clinical data analysis (n=217) and literature review (n=19 models) provide external validation. However, the study is entirely simulation-based; no experimental data were generated. The clinical dataset is limited by the availability of in vitro transporter data, and the classification of substrate status is binary without quantitative activity. The thresholds are conditional on model assumptions (e.g., linear kinetics, passive-only reabsorption), which may limit generalizability to drugs with active reabsorption or saturable transport.

Key Findings

The simulations demonstrate that basal uptake transporters are indispensable for net renal secretion (CR > 1.5). When basal uptake is absent, no virtual drug achieves net secretion, regardless of apical efflux or basal efflux activity. Five quantitative thresholds are identified: Pd > 5.5 L/h/m² (neutral, fu_b=1), PS_{act}/PS_{inf}(B) > 0.94 (non-cationic), PS_{act},influx(B) > 4.2 L/h, CL_{sec},Index > 14.4, and CL_{sec},App > 4.15 L/h. These thresholds are robust to fu_b (0.01) with minor adjustments. Clinical data show that 64% of 110 secretory drugs are substrates of OAT1/3 or OCT2, and only 6 are negative for all. Published MechKiM-PBPK models for 19 secretory drugs attribute >99% of basal uptake to transporters (except morphine at 70%). The study also explains why apical efflux alone fails: proximal secretion is always counterbalanced by distal reabsorption, as shown by the inverse relationship between FPT and FDT/CD.

🔗 Clinical & Regulatory Implications

The findings provide quantitative, in vitro-translatable criteria for anticipating net renal secretion and transporter-mediated drug–drug interaction (DDI) liability. Compounds meeting the thresholds (e.g., $PS_{act}/PS_{inf}(B) > 0.94$ for non-cationics, $PS_{act, influx}(B) > 4.2$ L/h) are likely to exhibit net secretion and may require clinical DDI studies with OAT/OCT inhibitors. The criteria also support mechanistic PBPK model development, enabling more accurate prediction of renal clearance and dose adjustments in special populations (e.g., renal impairment).

Strengths & Limitations

Strengths

- Systematic exploration of a large virtual drug space (30,000 compounds) covering a wide range of physicochemical and transporter parameters.
- Identification of universal, quantitative thresholds for net renal secretion that are directly translatable to in vitro assays.
- Integration of simulation results with clinical data and published PBPK models, providing external validation.
- Clear mechanistic explanation for why apical efflux alone is insufficient, resolving a long-standing question in renal pharmacology.
- The model code is provided as Supporting Information, enhancing reproducibility.
- Sensitivity analyses (fu.b, ionized permeability, distal tubule transporters) demonstrate robustness of the main conclusions.

Limitations (Acknowledged by Authors)

- Only neutral, mono-cationic, and mono-anionic compounds were considered; zwitterions and di-cationic/anionic compounds were not evaluated.
- Transporters were placed only in the proximal tubule; distal tubule/collecting duct transporters were not included in the main analysis (though a sensitivity analysis was performed).
- Each nephron segment is represented as a single compartment, which may underestimate reabsorption for highly permeable drugs.
- Active reabsorption (e.g., megalin, PEPT2) was not considered, though this would only strengthen the necessary conditions.
- Albumin-mediated uptake enhancement (AMUE) was not incorporated, which may affect highly protein-bound drugs.
- The ionized form permeability was assumed to be 1% of the unionized form; sensitivity analysis showed this assumption does not alter conclusions.

Limitations (Expert Review)

- The thresholds are derived from simulations with linear (non-saturable) transporter kinetics; in reality, saturation may occur at clinical concentrations, potentially altering the criteria.
- The model does not account for regional heterogeneity in transporter expression along the proximal tubule, which could affect the effective PS values.

- The clinical dataset relies on binary in vitro substrate classifications, which may not reflect quantitative transporter activity; drugs with weak substrate activity might be misclassified.
- The literature review of MechKiM-PBPK models is limited to published models, which may have selection bias.
- The study does not provide a direct experimental validation of the proposed thresholds; future work should test them in vitro or in vivo.

Generalizability

The findings are generalizable to small-molecule drugs that are passively reabsorbed and have linear transporter kinetics. The thresholds are derived for neutral, monoanionic, and monocationic compounds; zwitterions and di-cationic/anionic compounds may require different criteria. The model assumes no active reabsorption, which is reasonable for most drugs but may not apply to those with carrier-mediated reabsorption (e.g., peptides). The conclusions are consistent with clinical and PBPK data, supporting broad applicability.

Key Equations

Beta_{sec} (secretion fraction)

$$\beta_{\text{sec}} = \frac{\text{PS}_{\text{difeff,PT,CEtoPT}} + \text{PS}_{\text{act,efflux(A)}}}{\text{PS}_{\text{difeff,PT,CEtoVE}} + \text{PS}_{\text{difeff,PT,CEtoPT}} + \text{PS}_{\text{act,efflux(A)}} + \text{PS}_{\text{act,efflux(B)}}}$$

Fraction of drug secreted from proximal tubule cells to lumen relative to total efflux from cells, determining the direction of net transcellular flux.

Apparent reabsorption clearance

$$\text{CL}_{\text{reab,App}} = \text{PS}_{\text{difeff,PT,PTtoCE}} \cdot (1 - \beta_{\text{sec}})$$

Apparent reabsorption clearance in the proximal tubule, accounting for the fraction of drug that is not secreted back into the lumen.

Apparent secretion clearance

$$\text{CL}_{\text{sec,App}} = (\text{PS}_{\text{difeff,PT,VEtoCE}} + \text{PS}_{\text{act,influx(B)}}) \cdot \beta_{\text{sec}}$$

Apparent secretion clearance from blood to proximal tubule lumen, driven by passive diffusion and basal uptake transporters.

Secretion index

$$\text{CL}_{\text{sec,Index}} = \frac{\text{CL}_{\text{sec,App}}}{\text{CL}_{\text{reab,App}}}$$

Index of transcellular secretion efficiency, analogous to an efflux ratio across the proximal tubule epithelium.

Proximal tubule mass balance

$$\frac{dC_{PT}}{dt} = \frac{1}{V_{PT}} \{ C_b \cdot (f_{u,b} \text{ GFR} + f_{u,b} \text{ CL}_{\text{sec,App}}) - C_{PT} \text{ CL}_{\text{reab,App}} - C_{PT} Q_{u, PT} \}$$

Differential equation for drug concentration in the proximal tubule lumen, balancing filtration, secretion, reabsorption, and outflow.

Distal tubule mass balance

$$\frac{dC_{DT}}{dt} = \frac{1}{V_{DT}} \left\{ C_{PT} Q_{u, PT} - C_{DT} \left(\frac{P_d}{2} \text{SA}_{DT} + Q_{u, DT} \right) \right\}$$

Differential equation for drug concentration in the distal tubule lumen, accounting for inflow from PT and passive reabsorption.

Renal clearance (CL_{r,b})

$$\begin{aligned} \text{CL}_{r,b} &\approx f_{u,b} \text{ GFR} + \frac{C_{PT} \text{SA}_{PT} - C_{DT} \text{SA}_{DT}}{C_{DT}} + \frac{C_{PT} Q_{u, PT} - C_{DT} Q_{u, DT}}{C_{DT}} \\ &= \frac{C_{PT} \text{SA}_{PT} - C_{DT} \text{SA}_{DT}}{C_{DT}} + \frac{C_{PT} Q_{u, PT} - C_{DT} Q_{u, DT}}{C_{DT}} \end{aligned}$$

Final expression for blood-referenced renal clearance, showing the contributions of filtration and transporter-mediated secretion with downstream reabsorption factors.

Passive diffusion function

$$f(P_d) = 0.0728 P_d + \frac{1.25}{P_d} + 0.671$$

The function $f(P_d)$ that determines the maximum contribution of passive diffusion to net secretion; its minimum value defines the upper limit of the secretion term.

Figures & Tables

- **Figure 1:** Schematic of the MechKiM model showing glomerulus, proximal tubule (PT), distal tubule (DT), and collecting duct (CD) with blood, epithelial cell, and lumen compartments. Transporters are localized to the basal (uptake/efflux) and apical (efflux) membranes of PT cells.
 - *Significance:* Provides the structural basis for all simulations, illustrating the compartmental layout and transporter localization.
- **Figure 2:** (A) Boxplots of simulated CL_{r,b} for 30,000 virtual drugs under eight transporter knockout scenarios. (B) Clinical CL_{r,p} data stratified by in vitro uptake transporter status (positive, negative, or not determined).
 - *Significance:* Demonstrates that net secretion (CR > 1.5) only occurs when basal uptake transporters are present, and that clinical data align with this finding.

- **Figure 3:** Scatter plots showing relationships between key parameters (Pd, PSact/PSinf(B), PSact,influx(B), CLsec,Index, CLsec,App) and simulated CLr,b, with threshold boundaries indicated.
 - *Significance:* Identifies the quantitative thresholds that separate secretion-type from non-secretion-type compounds.
 - **Figure 4:** Plots of proximal tubule availability (FPT) versus Pd and distal availability (FDT,CD) versus Pd, and FPT versus FDT,CD, for the scenario with only apical efflux transporters.
 - *Significance:* Explains why apical efflux alone cannot yield net secretion: high Pd-driven proximal secretion is always accompanied by extensive distal reabsorption.
 - **Table 1:** Parameter ranges for virtual drug generation (Pd, PSact,influx(B), PSact,efflux(B), PSact,efflux(A), LRrenal, pKa).
 - *Significance:* Defines the explored parameter space, ensuring broad coverage of compound characteristics.
 - **Table 2:** Threshold values for Pd, PSact/PSinf(B), PSact,influx(B), CLsec,Index, and CLsec,App for neutral, anion, and cation classes at fu,b = 1 and fu,b = 0.01.
 - *Significance:* Provides the key quantitative criteria for net renal secretion, which are the main actionable outputs of the study.
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Code & Reproducibility Assessment

Model code is provided as Supporting Information (psp470336-sup-0004-supinfo.docx). The virtual drug parameter generation and simulation conditions are described in detail, allowing replication.

Supplementary Materials

Supporting information includes: (1) Data S1 (xlsx) with clinical CLr,p and in vitro transporter data; (2) Supporting Information (xlsx) with details of MechKiM-PBPK model parameters; (3) Supinfo (docx) containing model code and additional simulation results (Figures S1–S9, Tables S1–S4).

Future Directions

Future work should extend the analysis to zwitterions and di-cationic/anionic compounds, incorporate active reabsorption mechanisms (e.g., megalin, PEPT2), and include transporters in distal tubule/collecting duct segments. The impact of albumin-mediated uptake enhancement (AMUE) on highly protein-bound drugs warrants investigation. Experimental validation of the proposed in vitro thresholds using RPTEC or iPSC-derived proximal tubule cells would strengthen translation. Additionally, the criteria could be integrated into PBPK platforms to automate renal secretion prediction.

Expert Commentary

This paper addresses a long-standing gap in renal pharmacokinetics: why some drugs show net secretion while others do not, despite similar in vitro transporter profiles. The use of a mechanistic kidney model to systematically explore parameter space is a powerful approach, and the derived thresholds are actionable. The finding that apical efflux alone is insufficient is a critical insight, explaining why P-gp substrates often do not exhibit net secretion. The integration with clinical and PBPK data strengthens the conclusions. A key limitation is the assumption of linear kinetics and the exclusion of active reabsorption, but the authors appropriately acknowledge these. The proposed in vitro criteria (e.g., efflux ratio, PSact/PSinf) are practical and could be adopted in drug discovery. This work will likely influence how renal secretion potential is assessed and how PBPK models are built for renally cleared drugs.

Bottom Line

Net renal secretion ($CR > 1.5$) cannot occur without a substantial contribution of basal uptake transporters (e.g., OAT1/3, OCT2) in the proximal tubule. Mechanistic simulations define quantitative thresholds—such as transporter-mediated uptake contributing $>94\%$ of total basal uptake and $PS_{act,influx(B)} > 4.2$ L/h—that are consistent with clinical and PBPK data. These criteria are directly translatable to in vitro assays and can guide compound prioritization and DDI risk assessment.

Fact-check corrections

☒ Figures

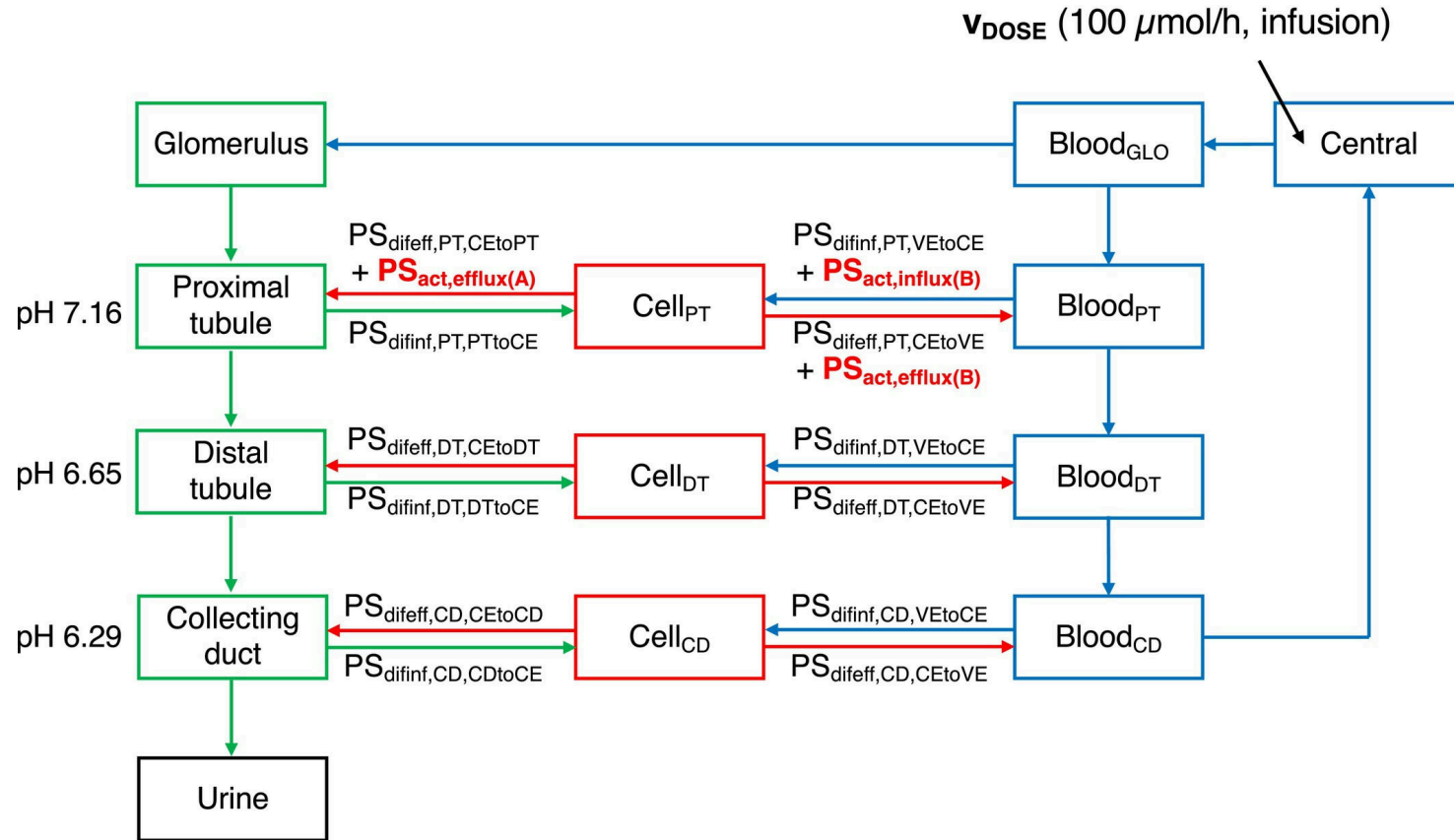


Figure 1: Model structure of the mechanistic kidney model (MechKiM). Physiological parameters are shown in TableS1. Transporters were distributed only in the proximal tubu

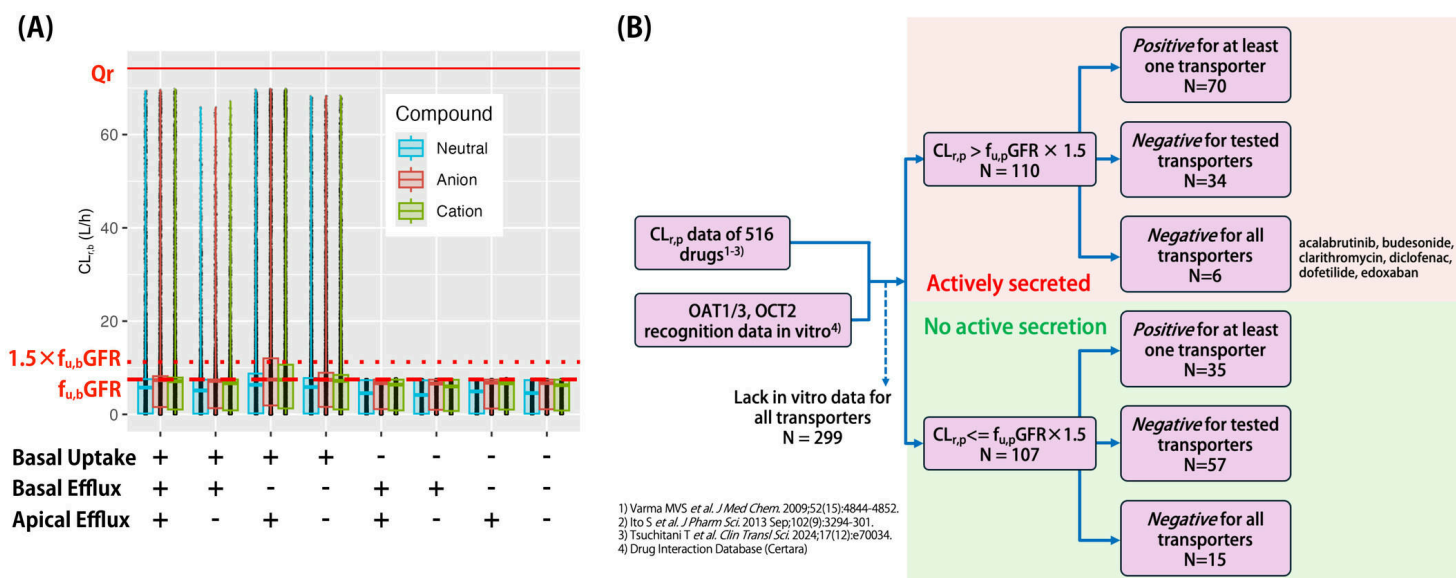
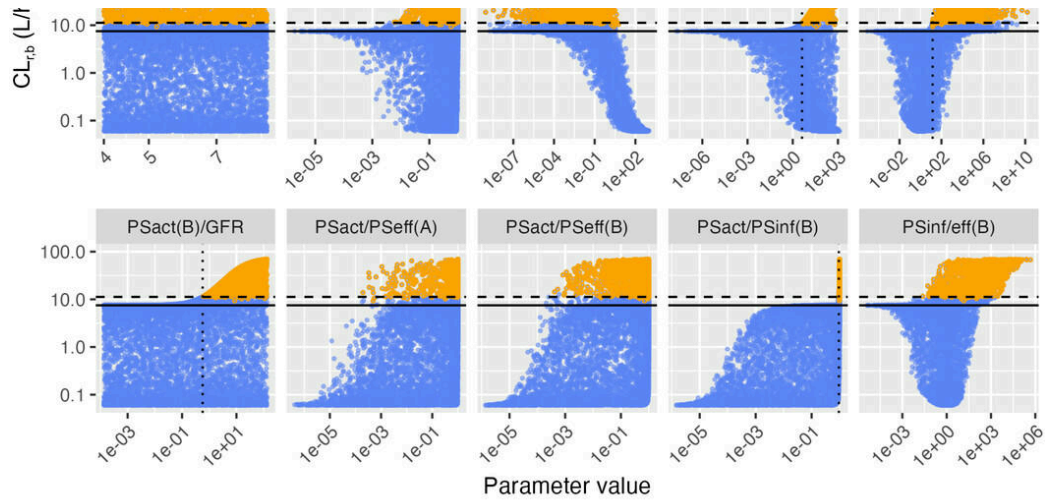
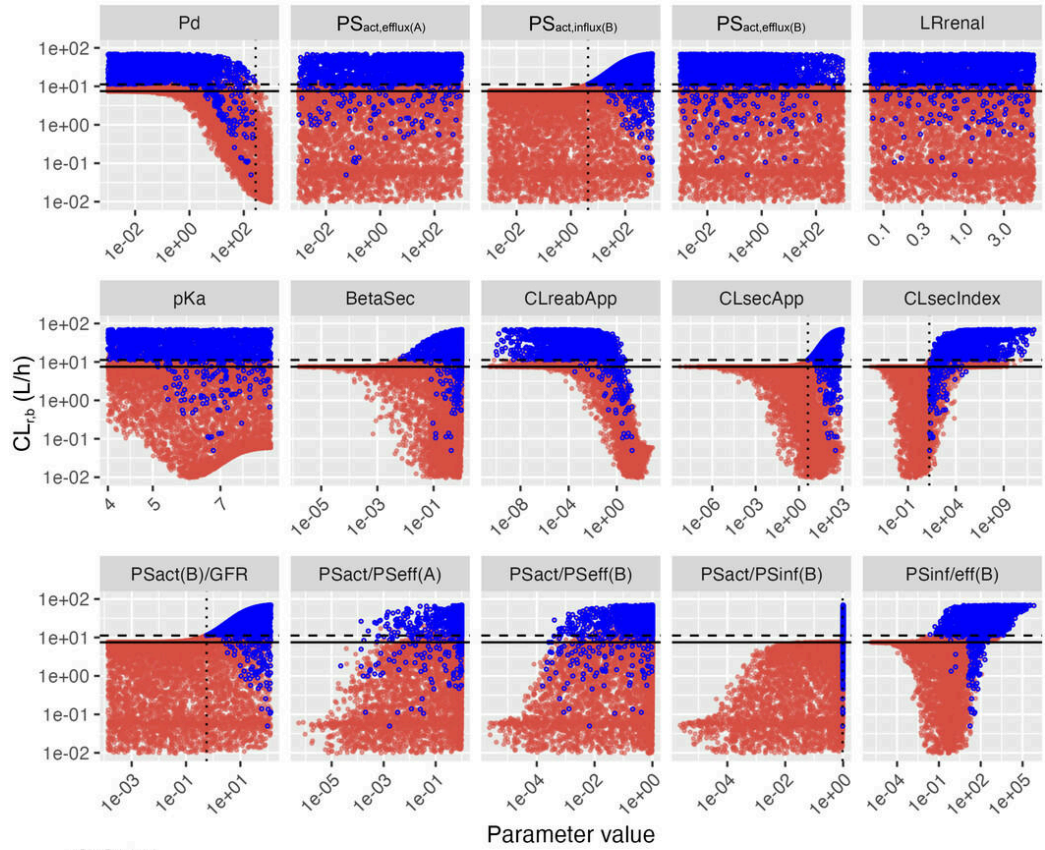


Figure 2: (A) Boxplot of simulated CL_{r,b} values under different transporter distribution scenarios using the MechKiM and (B) relationship between CL_{r,p} and in vitro transporter recognition data



(B) Anion



(C) Cation

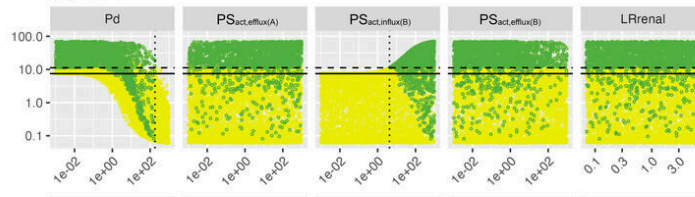


Figure 3: Relationship between parameter values and the simulated $CL_{r,b}$ values in the presence of basal uptake/efflux and apical efflux transporters. Each dot represents an

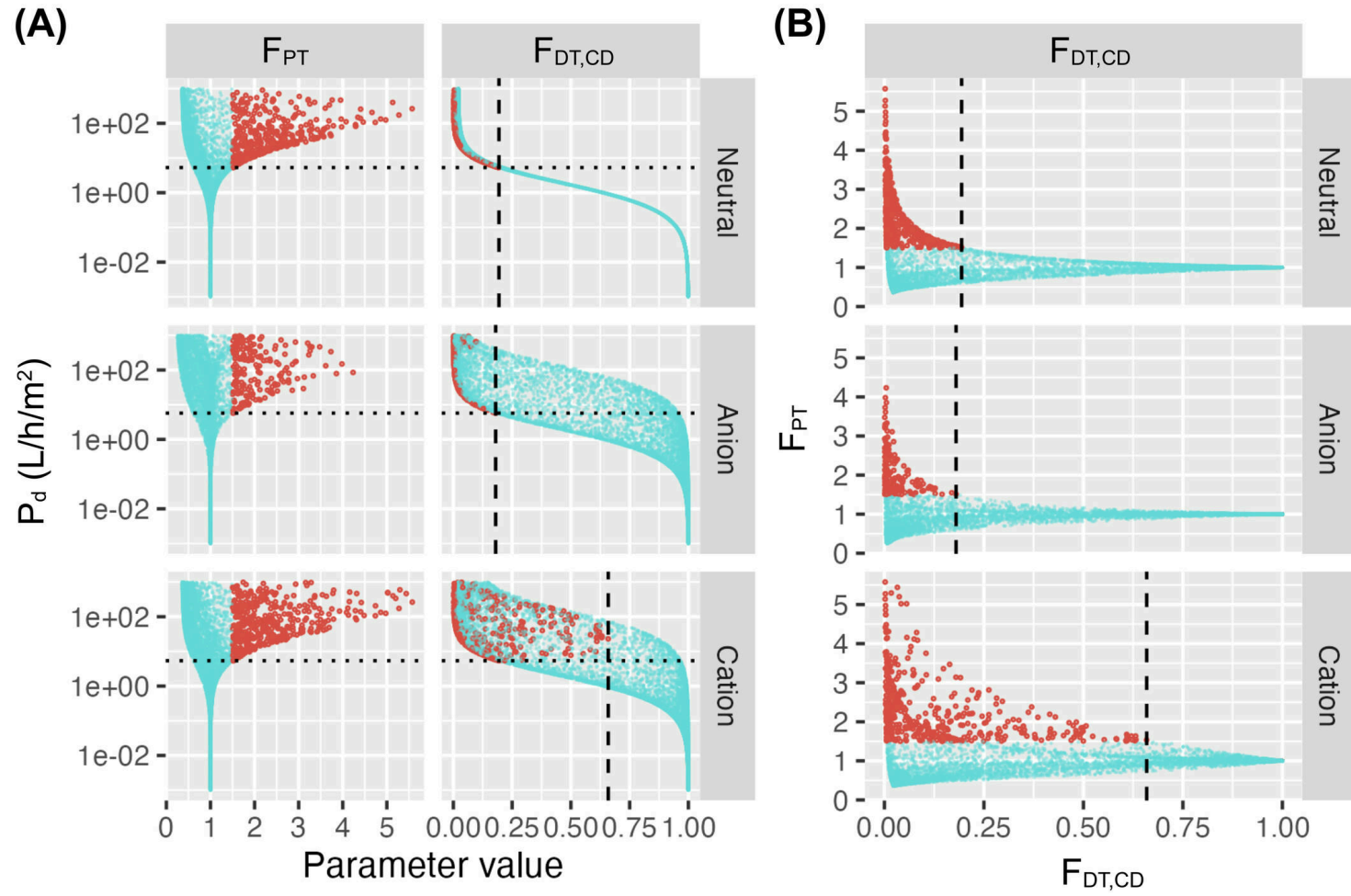


Figure 4: Relationship of “availabilities” of proximal and distal segments of the nephron tubules after basal uptake/efflux transporter knockout. The “availability” of the